

2-Month Dev Roadmap For 3rd Year Internships 🚀

You are at a VERY good stage right now.

You already:

- understand backend fundamentals
- think about systems instead of tutorials
- know Python
- know Flask/FastAPI basics
- understand async/workers/GIL at conceptual level

Now the goal is NOT:

learn everything

The goal is:

become internship-dangerous

Meaning:

- can build real backend systems
- can deploy them
- can explain architecture
- can clear decent interviews
- has strong projects

MOST IMPORTANT PRIORITY

You should optimize for:

projects + backend depth + deployment

NOT:

- endless theory
- 20 random frameworks
- tutorial hopping
- Kubernetes rabbit hole

YOUR BEST STACK

Do NOT split attention too much.

Master THIS:

```
FastAPI
PostgreSQL
Redis
JWT
Async
Docker
```

This stack has:

- startup demand
- backend internship demand
- AI infra relevance
- freelance relevance
- SaaS relevance

Huge ROI.

THE 2-MONTH PLAN

MONTH 1 → BUILD CORE BACKEND STRENGTH

WEEK 1 → FastAPI Properly

Learn

- FastAPI routes
- APIRouter
- Pydantic
- request/response models
- middleware
- async routes
- error handling
- dependency injection

Build

Mini Project

JWT Auth Backend

Features:

- signup
- login
- JWT access token
- refresh token
- protected routes
- role system

VERY IMPORTANT

Do NOT stop at CRUD.

Understand:

- why JWT stateless
- why refresh token exists
- why password hashing needed

WEEK 2 → PostgreSQL + Redis

Learn PostgreSQL

Learn:

- joins
- indexes
- foreign keys
- transactions
- query optimization basics

Use:

asyncpg

OR SQLAlchemy async.

Learn Redis

Learn:

- caching
- rate limiting
- OTP storage
- session storage
- pub/sub basics

Build

Add Redis to auth backend:

Features

- OTP login
- rate limiter
- token blacklist
- caching user profile

This instantly makes project look MUCH more advanced.

WEEK 3 → Async + Background Systems

Learn

- asyncio basics
- async DB calls
- background tasks
- websocket basics
- queues basics

Build

Realtime Notification System

Features:

- websocket notifications

- online user tracking
- Redis pub/sub
- background task processing

This project teaches REAL backend engineering.

WEEK 4 → Deployment + Docker

Learn

- Docker basics
 - docker-compose
 - Nginx basics
 - Linux basics
 - environment variables
 - VPS deployment
-

Deploy Your Projects

Deploy on:

- Render
 - Railway
 - EC2/VPS later
-

IMPORTANT

A deployed backend project >>> 10 unfinished local projects.

MONTH 2 → MAKE YOUR PROFILE STRONG

WEEK 5 → Build ONE Serious Project

Choose ONE.

Option 1 (BEST FOR YOU)

AI SaaS Backend

Example:

- PDF chatbot
- AI notes summarizer
- AI recommendation API
- AI automation backend

Use:

- FastAPI
- Redis
- PostgreSQL
- background tasks
- vector DB

This aligns perfectly with your existing AI chatbot experience.

Option 2

Telegram Automation Backend

Features:

- webhook handling
- scheduling
- admin dashboard API
- analytics
- Redis queues

VERY freelance-friendly.

Option 3

Realtime Collaboration Backend

Features:

- websocket
- room system
- Redis pub/sub
- presence tracking

Very strong backend project.

WEEK 6 → DSA Recovery

IMPORTANT.

Do NOT neglect DSA.

You do NOT need CP wizard level.

You need:

- solid patterns
- interview consistency

Focus:

NeetCode 150

Priority:

- arrays
- hashmap
- sliding window
- binary search
- stack
- linked list
- trees
- heap
- graphs basics

Target:

2-4 quality problems daily

NOT 15 random brute-force questions.

WEEK 7 → Resume + GitHub Polish

Resume

Backend-heavy resume.

Highlight:

- async architecture
- Redis
- JWT
- websocket
- deployment
- Docker
- PostgreSQL

Most students only show:

MERN todo app

You can stand out heavily.

GitHub

Your repos should have:

- README
 - architecture explanation
 - screenshots
 - API docs
 - deployment link
 - env example
-

WEEK 8 → Interview Simulation

Practice Explaining:

- async vs sync
- workers
- JWT
- Redis

- DB pooling
- why FastAPI
- project architecture
- scaling decisions

This matters MASSIVELY.

A lot of students can code. Few can explain systems.

WHAT YOU SHOULD IGNORE FOR NOW ❌

Do NOT waste these 2 months on:

- Kubernetes deep dive
- microservices obsession
- advanced distributed systems
- blockchain
- random frontend frameworks
- DevOps rabbit holes
- system design theory overload

Low ROI currently.

MOST IMPORTANT SKILL FOR INTERNSHIPS

Not:

knows 18 frameworks

But:

can build and explain one strong backend system

That wins.

THE INTERVIEW SUPERPOWER YOU ARE BUILDING

Most students say:

I used FastAPI because it is fast.

You can explain:

- async I/O
- worker concurrency
- Redis caching
- DB pooling
- stateless auth
- horizontal scalability

That creates VERY different impression.

YOUR BEST POSSIBLE END STATE AFTER 2 MONTHS

By end of break, ideally you should have:

Skills

- FastAPI
 - PostgreSQL
 - Redis
 - JWT
 - async concepts
 - Docker basics
 - deployment
-

Projects

At least:

1 serious backend project

AND

1 deployed project

DSA

Comfortable with:

- common patterns
 - medium-level interview questions
-

Resume

A backend-focused resume that looks MUCH stronger than average college profiles.

That combination is very strong for:

- startups
- backend internships
- AI startups
- freelance work
- remote internships

Huge ROI path.